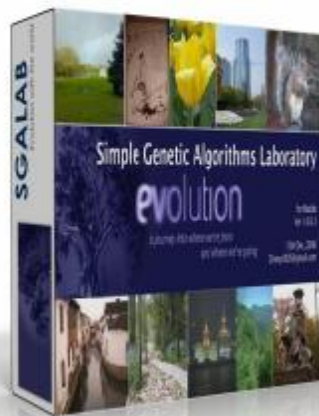




SxLAB

Evolution with the world



SxLAB Family

Chen Yi

chenyi2005@gmail.com
28th Feb., 2007

Contents

- „ What is SxLAB?
- „ SxLAB family members

What is SxLAB?

„ **SxLAB** is a family of Intelligent Algorithms and Dynamics Simulation toolbox for Matlab.

‘**S**,Simple’-- We will try our best to do things well , but can hardly do all the things.

‘**x** ’ -- We have a few of family members

‘**LAB**’ -- Laboratory , we are always in status of learning & evolution

SxLAB family members :: Intelligent Algorithm

Simple xxx Laboratory à

- „ SGALAB – ***Genetic Algorithms***
- „ SANTLAB – ***Ant Algorithms***
- „ SPSOLAB – ***Particle Swarm Optimization***
- „ SISLAB – ***Immune System***
- „ SSALAB – ***Simulated Annealing***

SxLAB family members :: Dynamics Simulation

Simple xxx Laboratory à

- „ SSTSLAB – *Space Tether Simulation & Control*
- „ SERALAB -- *Eigensystem realization Algorithm*

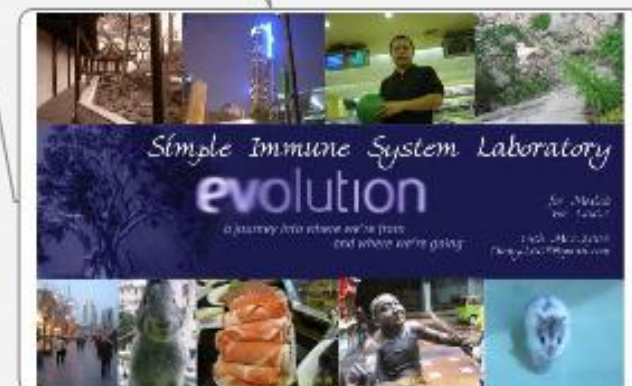
SSALAB = Simple Simulated Annealing Laboratory



SANTLAB = Simple ANT algorithm Laboratory



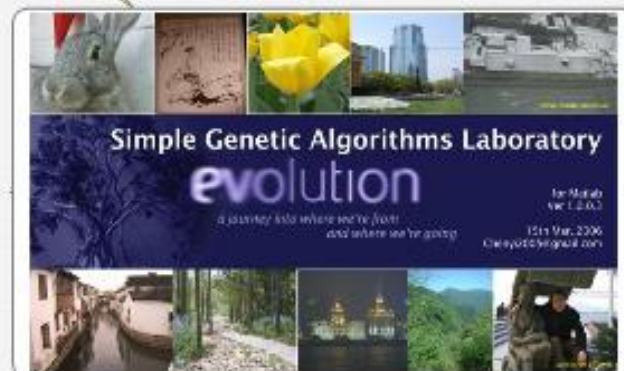
SISLAB = Simple Immune System Laboratory



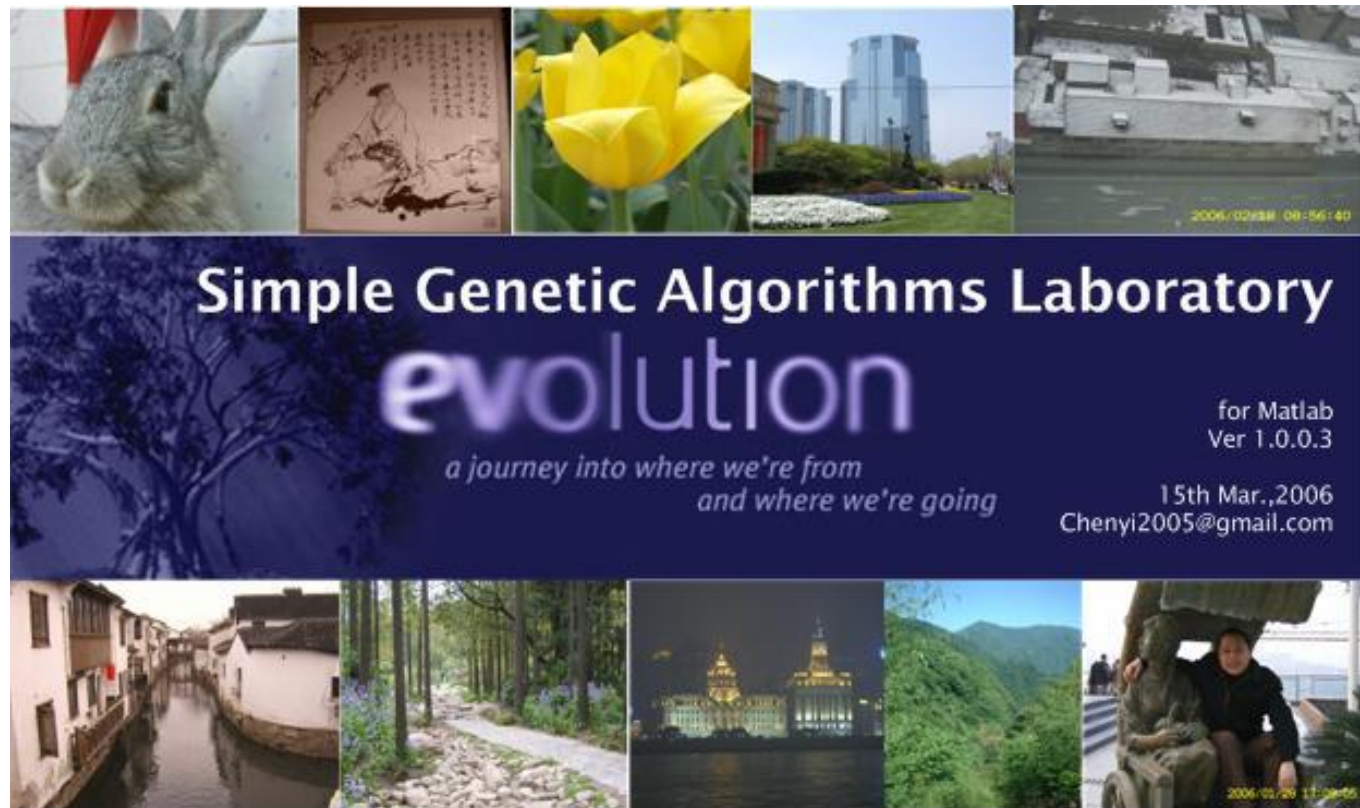
SPSOLAB = Simple Particle Swarm Optimization Laboratory



SGALAB = Simple Genetic Algorithms Laboratory



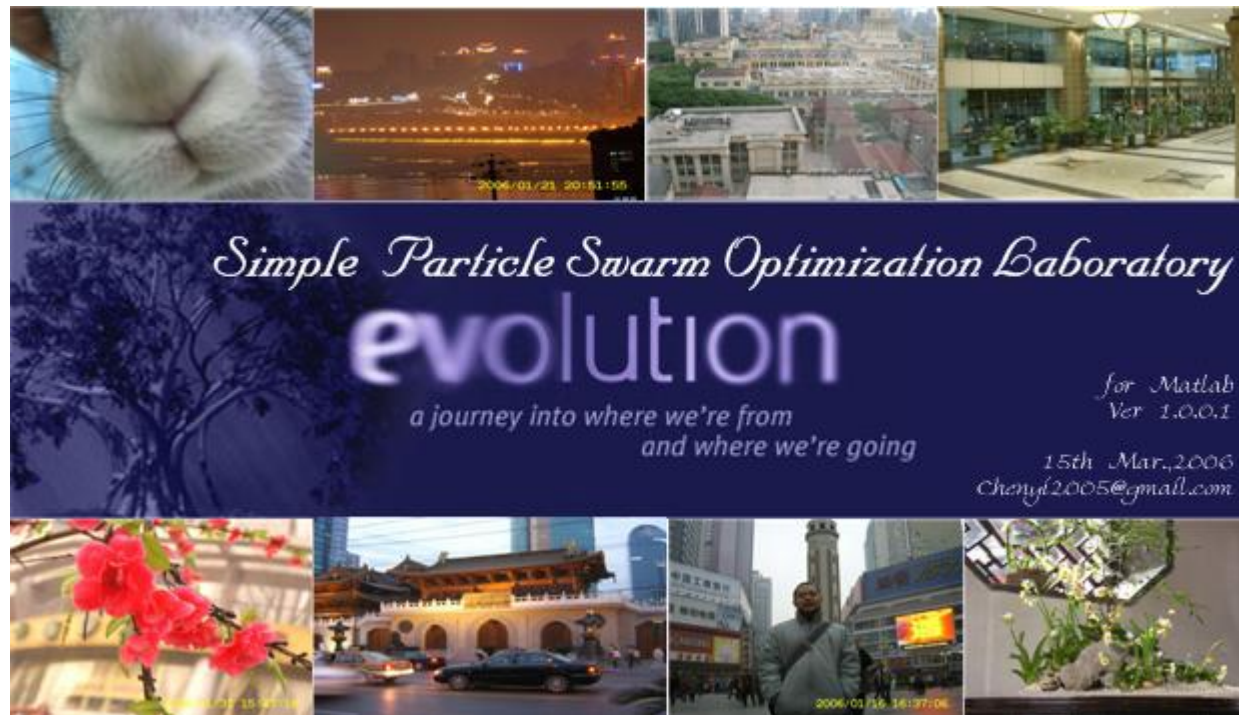
SGALAB



SISLAB



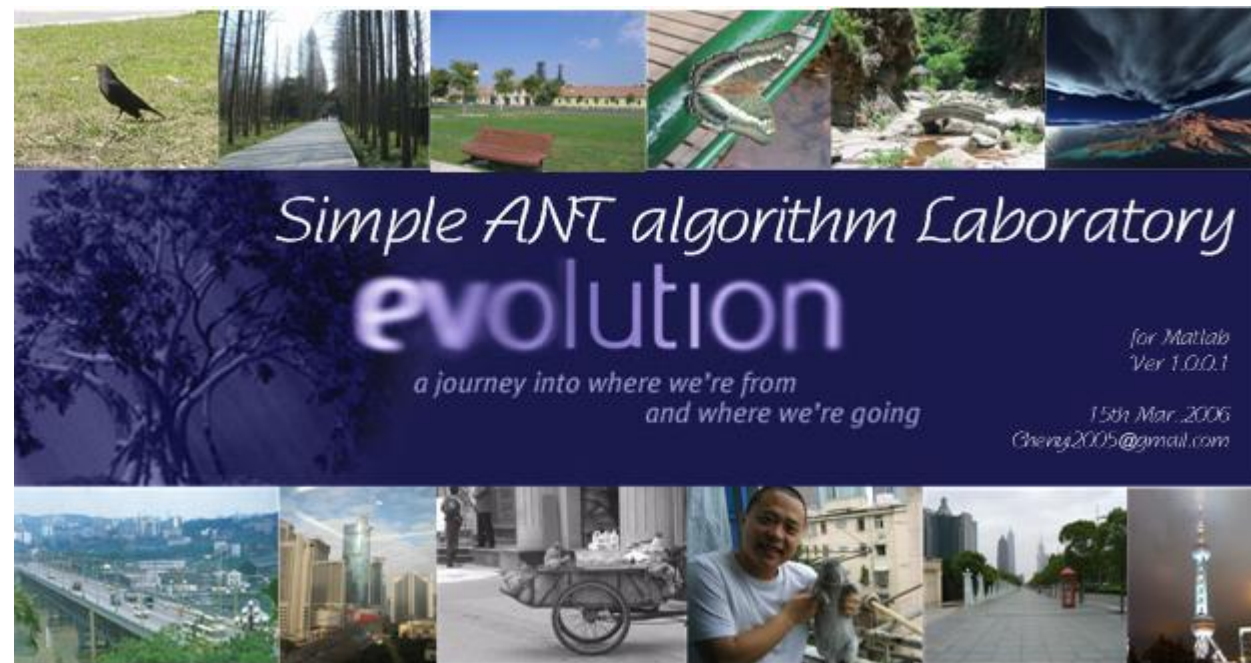
SPSOLAB



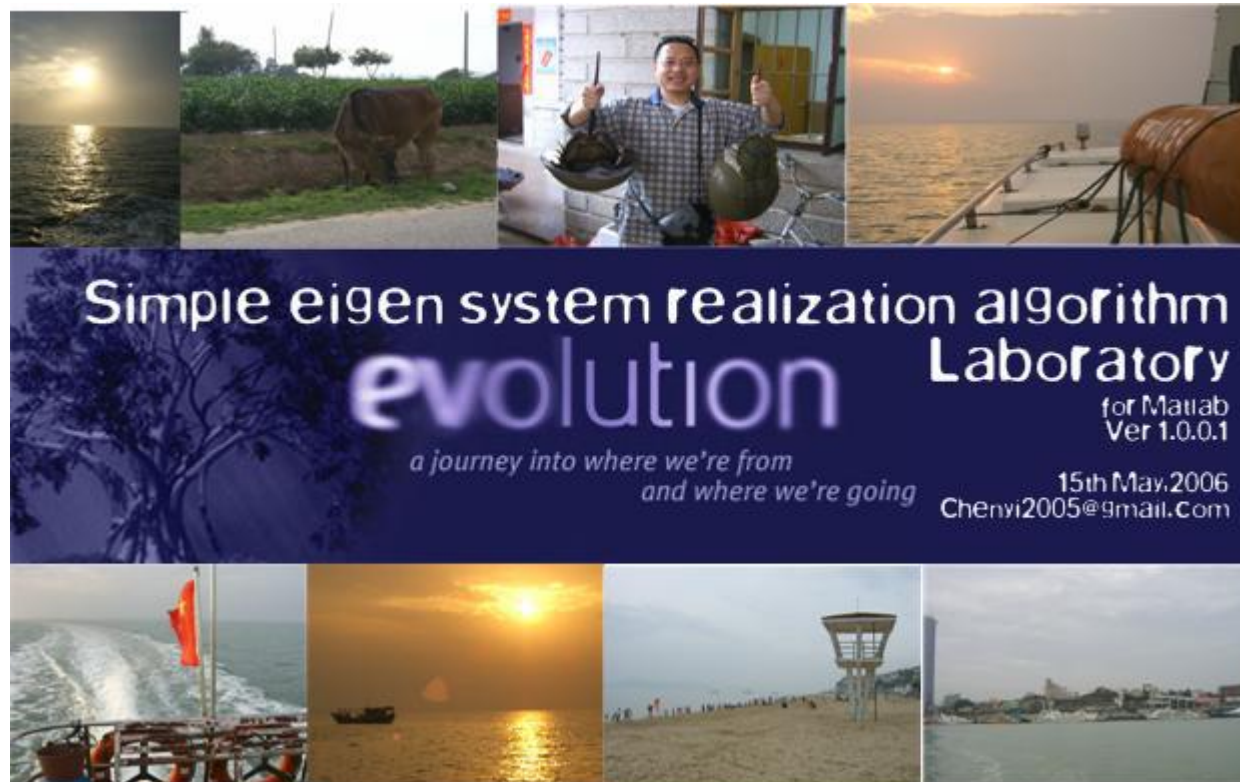
SSALAB



SANTLAB



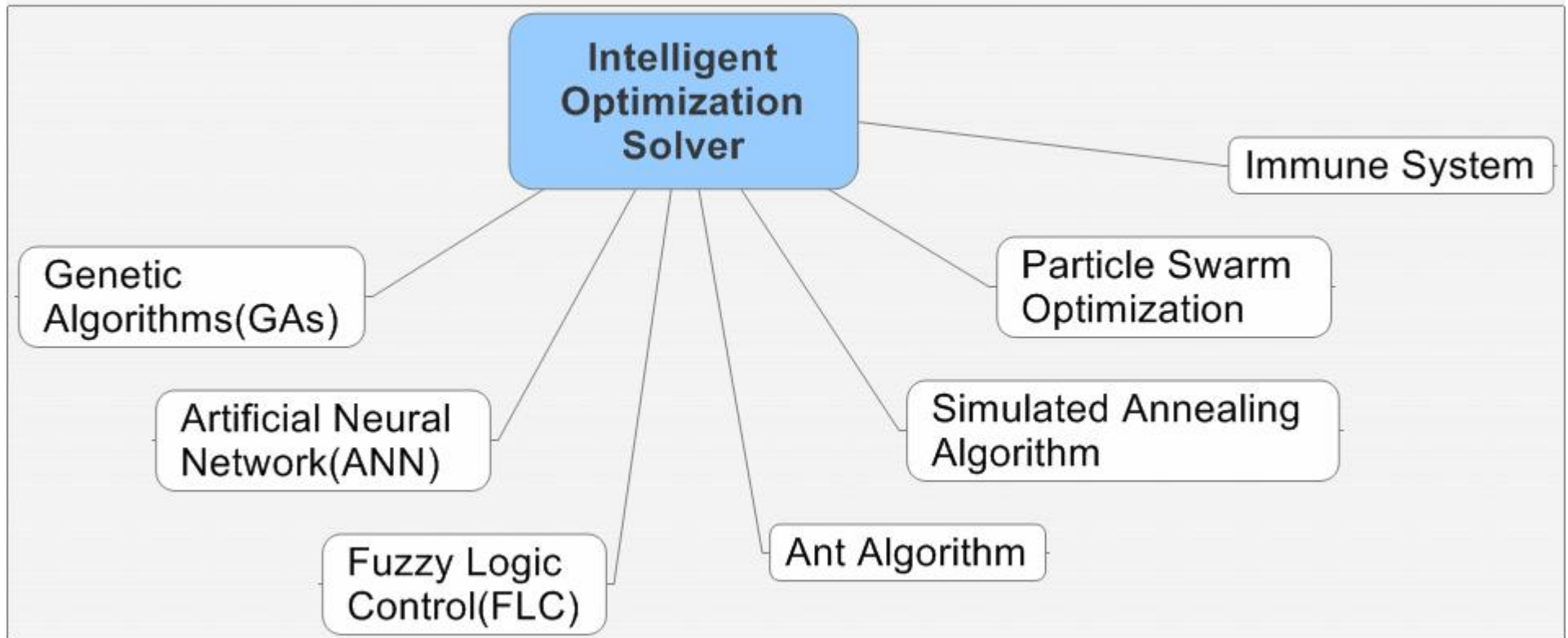
SERALAB



SSTSLAB



Add Intelligent algorithm in CAE Solver

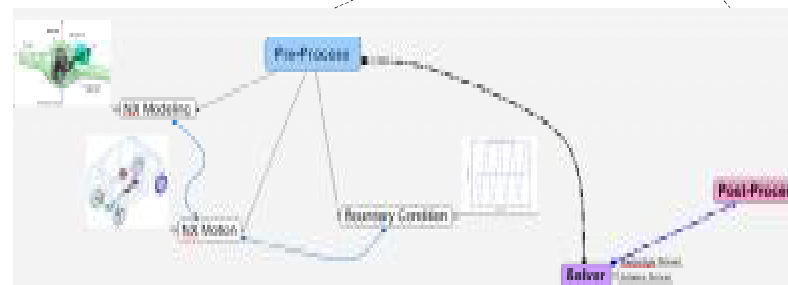
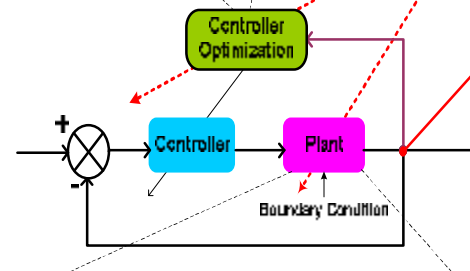
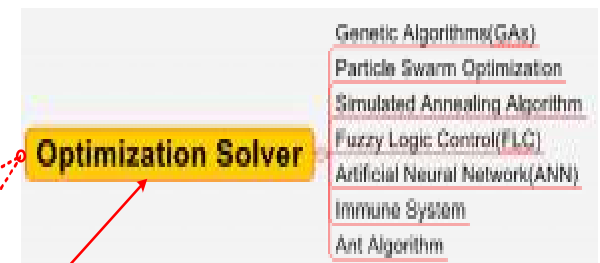
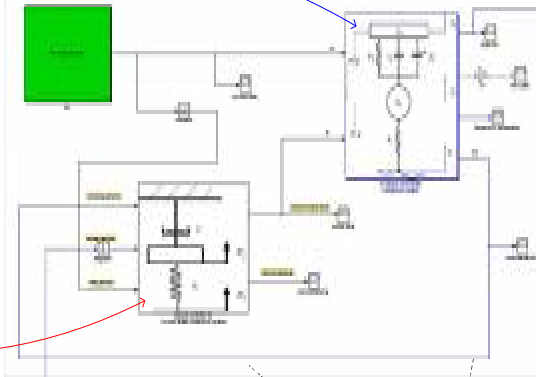
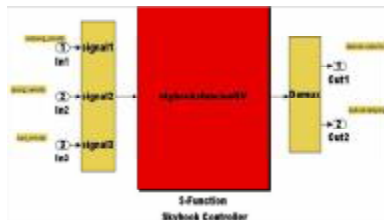
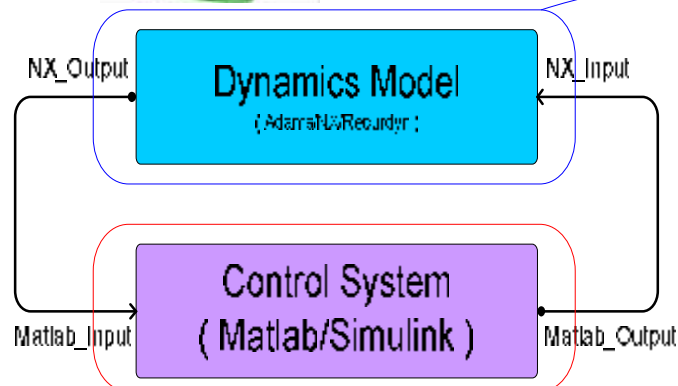


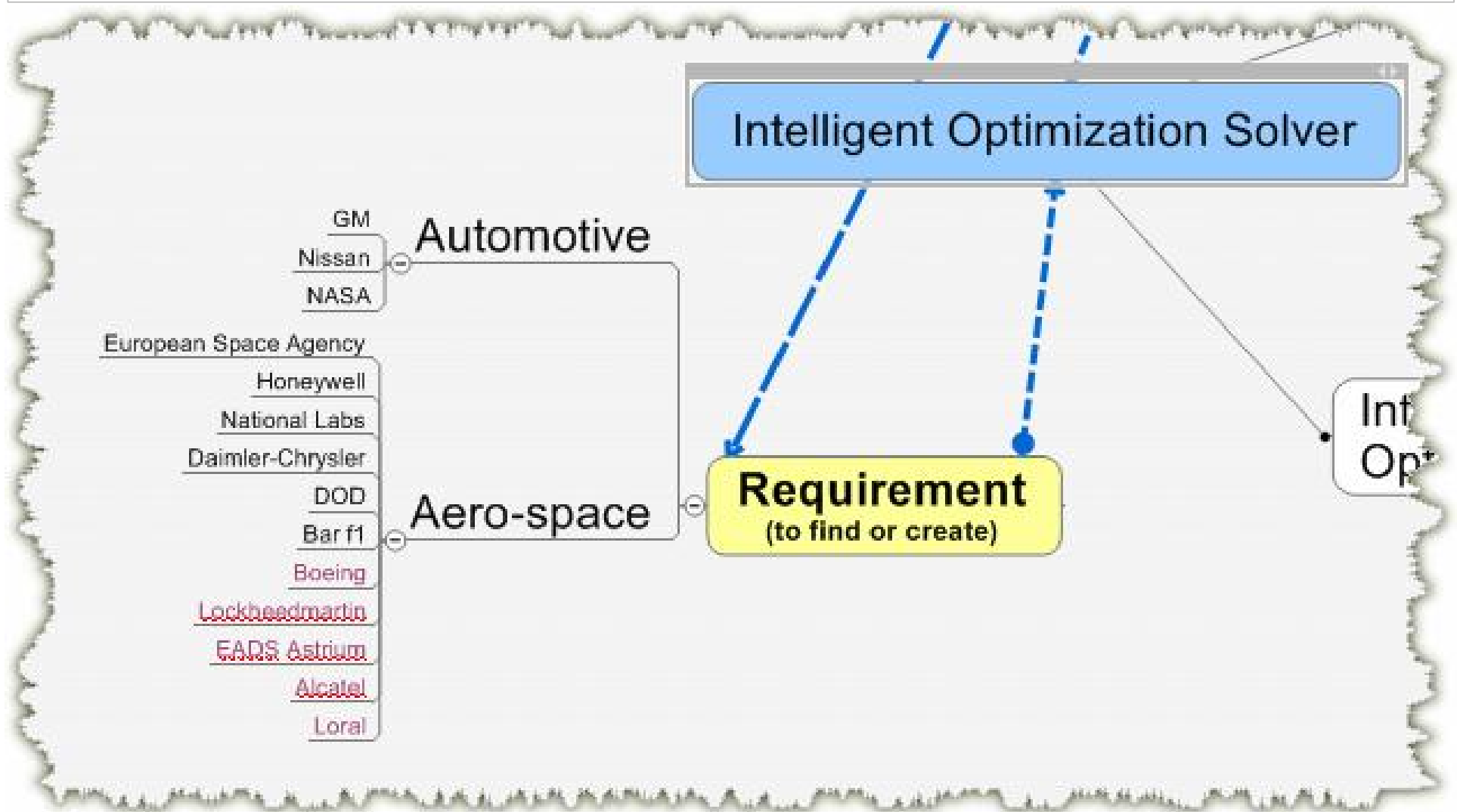
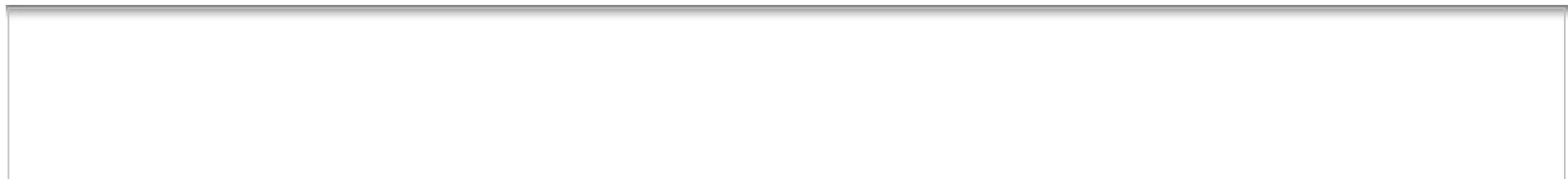


Step 1. we define NX_Input and NX_Output in NX, export as NX.mdl file;

Step 2. open the NX.mdl in matlab/simulink, add this mdl as plant for control system's

Step 3. 'Start' co-simulation in simulink, run matlab and NX so ver together







SxLAB

Evolution with the world



[临江仙]

陈昔注事益归去，
古道西风依旧。
人生长河水长东，
春去遇秋来，
天际寻鹄鸿。

日月星移话沧桑，
是非功过何在？
一路风雨兼程行，
唯清风明月，
独在情怀中。

